

Tuhin Chowdhury

Software Engineer

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EDUCATION

Sylhet Engineering College , affiliated of Shahjalal University of Science & Technology, Bachelor of Science in Computer Science & Engineering 09/2017 – 04/2022 | Sylhet , Bangladesh
Overall CGPA : 3.27 / 4.00 (Final Year CGPA : 3.7/4.00)
Thesis : An Automatic System for Identifying and Categorizing Tribal Clothing Based on Convolutional Neural Networks

SKILLS

Programming Languages

Java 21(Fluent),Python(Fluent), C++(Fluent), C(Fluent), C# (Basic)

Application Frameworks:

Spring Boot, FastAPI, .NET 8, Flutter, Android

AI, Machine Learning & Data Science:

MATLAB, PyTorch, TensorFlow, Scikit-Learn, LlamaIndex, Transformers, Pandas, Numpy, OpenCV, YOLO, HuggingFace

Tools & Platforms

AWS S3, PostgreSQL, Docker, JUnit, Linux, CI/CD, Git, GitLab, Jasper Reports, LaTeX, Overleaf

PUBLICATIONS

BLDAR: A Blending Ensemble Learning Approach for Primary Energy Consumption Analysis. (Book Chapter - 2025), Springer 📄

Implemented BLDAR, a novel blended ensemble model integrating LightGBM, Decision Tree, AdaBoost, and Random Forest regressors for forecasting primary energy consumption. The model achieved a state-of-the-art 90% R2 score, establishing a new benchmark for sustainable energy analytics.

An Automatic System for Identifying and Categorizing Tribal Clothing Based on Convolutional Neural Networks (2022), IEEE 📄

Developed a CNN-based system using the custom-made 'TribalBd' dataset to classify Bangladeshi tribal dresses, achieving 89.97% accuracy with YOLOv5, surpassing YOLOv6 (86.24%) and YOLOv7 (71.28%).

INDUSTRY EXPERIENCE

Software Engineer, Golden Harvest Infotech Limited 02/2023 – Present | Dhaka , Bangladesh

- Designed backend system architectures and built research-driven solutions, leading a small backend team, developing and optimizing microservices, collaborating with DevOps on CI/CD, and creating AI-based internal tools that supported 2,000+ daily users.
- Designed and developed a Nikon SDK-based imaging system for real-time capture and automated processing, while driving database optimization and contributing across the full Agile SDLC, demonstrating strong capability in applied system research.

Software Engineer(Intern), Rokkhi IT Limited 06/2022 – 01/2023 | Dhaka, Bangladesh

Implemented an automated document processing feature by building a Java service that connects to an EasyOCR Python module, reducing manual data entry by over 50%

PROJECTS

Hospital Management System ,(MHM, Houston , TX),

Tech Stack : SpringBoot 3, PostgreSQL, RAG, WebClient, Openai 4 , ChromaDB, Mcp, N8N, Betterstack, AWS S3, Docker.

- Built Spring Boot microservices to manage patient information, bed management, admissions, and cross-department workflows across OPD, IPD, IAM, and EHR.
- Implemented secure communication using JWT, timestamp validation, IPC-based internal service calls, and Betterstack for centralized logging.
- Optimized performance with PostgreSQL through custom SQL, indexing, and optimized joins, along with asynchronous processing, scheduled jobs, and reliable transactional operations.
- Developed N8N automation workflows, OCR-based patient registration, real-time bed allocation, nurse shift tracking, and duplicate detection, significantly improving system accuracy and operational efficiency.

Church Census Records Archiving Project , (Family Search , Utah),

Tech Stack : C# , .NET 8 , WPF , GraphQL ,SQLite , XAML , Python , Nikon SDK

- Engineered a high-throughput DSLR imaging system using C#, .NET 8, and Nikon SDK, delivering 1,000+ images per hour with real-time line-segmentation checks, clarity assessment, contrast analysis, and blur detection.
- Designed and built a dual-image capture workflow that saves two cropped images simultaneously, a capability with no existing online solution, ensuring accurate and consistent outputs for large-scale digitization.
- Led the backend and hardware system design, integrating real-time processing pipelines, optimized storage management, and a bulk sharing workflow that preserves image quality for millions of historical pages.

Satellite Frequency Renting and Auto Billing System (Bangladesh Satellite Corporation , Dhaka),

Tech Stack : Java , Spring Boot , JasperSoft , Sendgrid , PostgreSQL

- Built a Spring Boot billing management system for multiple clients to automate 100% of invoice, chalan, and payment notifications via SMTP email, handling 15+ JasperReports with multi-report and graph-based analytics, deployed using Docker.
- Calculated satellite frequency usage on a pay-per-use model, integrating dynamic reporting and analytics, which eliminated manual reporting and reduced operational effort by 50% while ensuring reliable, scalable billing workflows.

ACHIEVEMENTS & HOUNOURS

Selected for Bangladesh-Japan Engineers Training Program, Miyazaki University Competitively selected for an international training program organized by Miyazaki University in collaboration with the Bangladesh Government, focused on advanced engineering practices and cross-cultural technical exchange.	2022 Miyazaki , japan
Best IoT Project Recognition, University Course Project Awarded for developing an innovative Smart Home automation system.	2019 Sylhet, Bangladesh
First Prize, Intra-Batch Programming Contest, University Secured first place in competitive programming contest among batch peers.	2018 Sylhet , Bangladesh
First Prize, Essay Writing Competition, University Campus Secured first place in university-wide essay writing competition.	2017 Sylhet, Bangladesh
District Commissioner Scholarship Award, Academic Excellence 2016 Awarded scholarship for achieving GPA 5.0/5.0 in Higher Secondary Certificate examination.	2016 Chittagong , Bangladesh

TEACHING & MENTORSHIP

Java & Data Structures Instructor Mentored over 10+ students, resulting in improved programming skill and successful project completions.	
Instructor, Admission Academic Care Taught Mathematics and Physics to college-level students, delivering comprehensive lessons and supporting academic development.	Sylhet

PARTICIPATIONS

IEEE Computer Society Bangladesh Chapter Symposium Research abstract accepted for presentation.	11/2024
Fourth International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), IEEE Presented paper on Computer Vision	03/2022
Facebook Developer Circles Challenge META Participated in Chatbot-Developments Hackathons for Messenger Platform and placed in top 20 .	02/2021
ICPC Intra University Selection Programming Contest Our team ranked top 50 among 900 teams.	10/2019
Leading University National Programming Contest Team ranking was 72 out of 190 teams.	02/2018

EXTRACURRICULAR ACTIVITIES

Computer Society, General Secretary Organized the university's annual programming contest and coordinated departmental technical events.	11/2021 – 11/2022
Bohubrihi - Edtech Platform, Campus Director Organized programming sessions and workshops on behalf of Bohubrihi to enhance student skills.	01/2021 – 01/2022
SECPA -University Photography Club, Vice President & Instructor Served as Image Aesthetics Instructor and conducted workshops on visual composition and photography techniques. My work has won several national and international awards at photography exhibitions.	01/2020 – 01/2022
HULT Prize SEC, Founding Campus Director Organized online seminars and programs to promote social entrepreneurship on campus.	01/2020 – 01/2021

REFERENCES

- Ashraful Islam,** *PhD Candidate,* Department of Computer Science, University of Regina, Canada
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- Md. Ariful Islam,** *Chief Technical Officer & System Architect,* Golden Harvest Infotech Limited
Email : arif@inspi8bd.com